

Semi-meanders with a fixed winding deficit: the third and fourth diagonals of A400429

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Abstract

We prove Hunter Hogan's formulas for the third and fourth diagonals of OEIS A400429 by a finite arch transfer and exact rational certificates. The fourth diagonal has degree six and minimal period 24; we express it as a polynomial with separate corrections of periods two, three and eight. For every fixed j , the j th diagonal is eventually a quasipolynomial of degree $2j - 2$, with positive rational leading coefficient independent of the residue and period dividing $2 \operatorname{lcm}(1, \dots, j)$. The transfer gives independent exact data through $n = 10000$.

1 The two formulas

Place $2n$ vertices on a line and fix the lower rainbow matching $i \leftrightarrow 2n + 1 - i$. An upper noncrossing perfect matching defines a semi-meander when its union with the rainbow has one component [1]. Its winding number is the number of upper arches crossing the midpoint. In the notation of A400429 [2], set

$$D_j(n) = T(n, \lfloor n/2 \rfloor - j + 1), \quad n \geq 2j.$$

Thus $D_j(n)$ has winding $n - 2j$, and has exactly j upper arches entirely in each half of the line. The degree $2j - 2$ below concerns this indexing; j is the number of such arches in each half.

Theorem 1. *For every $n \geq 6$,*

$$D_3(n) = \frac{13n^4 + 24n^3 - 284n^2 - 4560n + 17920 + (n \bmod 2)(18n^2 - 72n + 189) - 64\mathbf{1}_{3|n}}{288}. \quad (1)$$

This quasipolynomial has degree four and minimal period six. For every $n \geq 8$,

$$D_4(n) = P(n) + (-1)^n E(n) + C_3(n) + R_8(n), \quad (2)$$

where

$$\begin{aligned} P(n) &= \frac{1}{138240} (294n^6 - 60n^5 - 3895n^4 - 232300n^3 \\ &\quad - 326739n^2 + 16955410n - 48570245), \\ E(n) &= \frac{-5n^4 + 44n^3 - 289n^2 + 358n + 4473}{3072}, \\ C_3(n) &= \frac{4}{27} \begin{cases} n - 2, & n \equiv 0 \pmod{3}, \\ 3, & n \equiv 1 \pmod{3}, \\ -n - 1, & n \equiv 2 \pmod{3}, \end{cases} \\ 16R_8(n) &= (3, -3, 5, -1, -5, 5, -3, -1)_{n \bmod 8}. \end{aligned}$$

The fourth diagonal has degree six, leading coefficient $49/23040$, and minimal period 24. Formula (2) equals Hogan's supplied period-24 Horner formula exactly.

The proof combines a combinatorial reduction valid for all n with finite exact calculations specified in Section 4. Consequently, Theorem 1 is a computer-assisted proof, rather than a conclusion drawn from agreement of initial terms.

2 A bounded arch transfer

Read the vertices in pairs from the outside towards the midpoint: at step t read vertex t and vertex $2n + 1 - t$ and join them by their lower rainbow edge. An upper arch wholly within a half is read using the usual stack rule. An opening puts its endpoint on that side's stack; a closing removes the top endpoint and joins it to the new vertex. An arch crossing the midpoint leaves one endpoint on each stack. At the end, the two stacks have equal length, and their endpoints are joined in corresponding order from bottom to top. This construction is bijective with the upper noncrossing matchings.

Let u, v be the numbers of closings already used on the left and right. Each must eventually equal j . Before any simplification, the two stack heights at step t are $t - 2u$ and $t - 2v$. Every component of the partial union is a path with two exposed endpoints, unless it has already become a cycle. We record the pairing of exposed endpoints determined by these paths.

Lemma 1. *Suppose that the retained stacks have lengths a, b . Put*

$$k = \max\{0, \min(a - (j - u), b - (j - v))\}.$$

The bottom k endpoints on each stack may be joined in corresponding order and removed immediately. This operation preserves every possible completion with j closings on each side.

Proof. At most $j - u$ left endpoints can still be popped. An existing endpoint with at least $j - u$ endpoints above it cannot be popped, even if no new endpoints are opened; future openings can only increase that depth. The same holds on the right. Thus all the indicated endpoints must remain until the final midpoint joining. Since the retained stacks have had the same number of bottom endpoints removed, corresponding bottom ranks still have their final counterparts. Their final edges can therefore be added now. Adding these prescribed ranks commutes with all later additions of edges. $\square$

After this operation $a - b = 2(v - u)$ and

$$a + b \leq 2(j - \min(u, v)) \leq 2j. \quad (3)$$

Indeed, if $v \geq u$, either $a \leq j - u$ or $b \leq j - v$; together with $a = b + 2(v - u)$ either inequality implies $a + b \leq 2(j - u)$. The other case is symmetric. There are therefore finitely many states

$$(u, v, a, p), \quad 0 \leq u, v \leq j,$$

where p pairs the $a + b$ retained endpoints. The pairing is noncrossing when the boundary is ordered as the left stack followed by the reversed right stack. The initial state is $(0, 0, 0, \emptyset)$.

There are four possible steps: opening or closing on each side. A closing requires a nonempty stack and an unused closing. Add the new lower edge and the prescribed upper edges, then perform Lemma 1. If a component becomes a cycle, reject the step, except when there is exactly one cycle, no exposed endpoint remains, and $u = v = j$. In that case accept and give this terminal state no outgoing steps. A closed component cannot subsequently be joined to any new vertex, so this rule is equivalent to final connectedness.

These rules are both necessary and sufficient. Stack operations enumerate every noncrossing upper matching once, the early edges are forced by Lemma 1, and the pairing retains exactly the information needed for all subsequent joins. Thus accepted words of length n are counted by $D_j(n)$.

3 Degree and period for general j

Theorem 2. *For every fixed integer $j \geq 1$, $D_j(n)$ is eventually a quasipolynomial of degree exactly $2j - 2$. Its leading coefficient c_j is positive, rational and independent of the residue. Its minimal eventual period divides*

$$L_j = 2 \operatorname{lcm}(1, 2, \dots, j).$$

For $j \geq 2$, all residue-dependent coefficients have degree at most $2j - 4$. The finite transfer determines both c_j and the true period. In particular,

$$c_1 = 2, \quad c_2 = \frac{1}{2}, \quad c_3 = \frac{13}{288}, \quad c_4 = \frac{49}{23040}.$$

Proof. Write O for opening on both sides. Every other step increases $u + v$. Thus every directed cycle of the transfer consists entirely of O steps and is contained in one level $u + v$.

Consider a sufficiently long surviving run of O steps at fixed u, v . The bound (3) becomes an equality: new openings are followed by joining and removing the bottom pair. If $v > u$, put $\delta = v - u$. The retained lengths are

$$a = j + v - 2u = 2\delta + b, \quad b = j - v.$$

After b further O steps, every right endpoint was created during this run and is paired with the left endpoint created at the same step. These b left-right pairs occupy the top b left endpoints. The bottom 2δ left endpoints are paired among themselves. The next O step removes the bottom left endpoint and the bottom right endpoint. The latter was paired to the first left endpoint above this 2δ -endpoint group. Joining their partners moves the first endpoint of the group to its last position. Thus O rotates its matching by one endpoint. Its period divides 2δ . The case $u > v$ is symmetric. When $u = v$, all retained pairs eventually join equal ranks, and the next bottom join closes a component; there is no surviving cycle.

Every cycle length therefore divides L_j . Decompose an accepting path into its finite passages and its repeated O cycles. The generating function is a finite sum of rational terms with denominators that are products of $1 - z^p$, where $p \mid L_j$. This proves eventual quasipolynomiality. The level $u + v = 0$ has no cycle: repeated O steps would close an isolated rainbow pair. The level $u + v = 2j$ has no nonterminal state, since all remaining endpoints are joined. An accepting path therefore meets at most $2j - 1$ cyclic levels. Every pole has order at most $2j - 1$, giving degree at most $2j - 2$.

To prove equality, use the following explicit states. For $v > u$ let $A_{u,v}^\epsilon$ have $b = j - v$ left-right pairs at the tops of its stacks, and an adjacent matching on its bottom $2(v - u)$ left endpoints. For $\epsilon = 0$ this matching is $(0, 1), (2, 3), \dots$; for $\epsilon = 1$ rotate it by one endpoint. The two phases coincide when $v - u = 1$; otherwise O interchanges them. Write R for left opening and right closing, and L for left closing and right opening. Reading words from left to right, the initial word ORO^{j-1} reaches $A_{0,1}^0$. For $1 \leq v < j$ put $q = j - v - 1$. From $A_{0,v}^{q \bmod 2}$, the word RO^q reaches $A_{0,v+1}^0$. To see this, label the left endpoints from zero and put $d = v$. The R step turns the last left-right pair into the adjacent pair $(2d + q, 2d + q + 1)$. During the next q joins, the removed bottom left endpoint t is connected through the removed right endpoint to the left endpoint $t + 2d$, for $t = 0, \dots, q - 1$. Thus the old matching on residues modulo $2d$ is transported to the window $q, \dots, q + 2d - 1$, followed by the new adjacent pair. The relative phase is $q - q = 0$ modulo two, as required.

From $A_{u,j}^0$, a L step reaches $A_{u+1,j}^1$ if $u < j - 1$. It removes the first and last endpoints and joins their former partners, which is the rotated adjacent matching. One O step returns to phase zero. From $A_{j-1,j}^0$, the L step accepts. These transitions exhibit an accepting path through a cycle at each of levels $1, 2, \dots, 2j - 1$.

For $j \geq 2$, insert arbitrary full turns at these cycles. The first and last cycles have length one and the other $2j - 3$ cycles may be traversed in two steps. The defect word is $\text{R}^j \text{L}^j$, so the inserted lengths determine distinct words. This subfamily has a generating function

$$\frac{z^{N_j}}{(1 - z)^2(1 - z^2)^{2j-3}}$$

for a fixed integer N_j . Its coefficient has positive leading term $n^{2j-2}/(2^{2j-3}(2j-2)!)$. Hence the degree cannot be smaller. For $j = 1$, direct enumeration gives $D_1(2) = 1$ and $D_1(n) = 2$ for $n \geq 3$.

Finally, cycles at levels 1 and $2j - 1$ have $|u - v| = 1$. Their two-endpoint matching has period one, so neither can contribute a pole away from 1. Such poles have order at most $2j - 3$. The leading coefficient is consequently constant across residues, and

$$c_j = \frac{1}{(2j-2)!} \lim_{z \rightarrow 1} (1 - z)^{2j-1} \sum_{n \geq 2j} D_j(n) z^n > 0.$$

It is rational because the generating function is rational over $\mathbb{Q}$. This also proves the stated bound on the periodic coefficients. $\square$

4 Exact generating functions and certificates

Let $F_j(z) = \sum_{n \geq 2j} D_j(n)z^n$. The transfer gives

$$F_3(z) = \frac{2z^6 Q_3(z)}{(1-z)^5(1+z)^3(1+z+z^2)}, \quad (4)$$

$$Q_3(z) = 4 + 17z + 17z^2 - 5z^3 - 22z^4 - 13z^5 + 6z^6 + 9z^7 + z^8 - z^9,$$

$$F_4(z) = \frac{2z^8 Q_4(z)}{(1-z)^7(1+z)^5(1+z^2)(1+z^4)(1+z+z^2)^2}, \quad (5)$$

where

$$\begin{aligned} Q_4(z) = & 21 + 131z + 292z^2 + 320z^3 + 118z^4 - 102z^5 - 89z^6 \\ & + 107z^7 + 193z^8 + 37z^9 - 196z^{10} - 226z^{11} - 8z^{12} \\ & + 217z^{13} + 199z^{14} + 5z^{15} - 106z^{16} - 58z^{17} \\ & + 14z^{18} + 19z^{19} - 2z^{20} - 4z^{21}. \end{aligned}$$

The reachable transfers have the following sizes and cycle lengths. The entries list distinct lengths, including the one-step cycles.

j	States	Transitions	O-cycle lengths
1	4	8	1
2	16	33	1,2
3	56	127	1,2,3
4	187	458	1,2,3,4,8

The rational certificates `certificate3.json` and `certificate4.json` establish (4) and (5). For every reachable state s , each certificate gives its pairing, all outgoing labelled transitions, and a rational function f_s . The terminal function is 1. The independent checker `verify.py` reconstructs each transition using connected components and checks, by multiplication of integer polynomials, every identity

$$f_s = z \sum_{s \rightarrow t} f_t. \quad (6)$$

All denominators have nonzero constant term. Thus these identities have a unique solution as formal power series: each coefficient of degree n is determined by coefficients of degree $n-1$. The initial-state functions in the certificates are precisely (4) and (5). Reachability and all four possible step choices are checked for every state; no state cutoff or guessed recurrence is used.

Summing (1) along its six residue classes gives (4). Summing Hogan's Horner formula along its 24 residue classes gives (5); this equality is checked with integer polynomials. Newton's finite-difference identity suffices because the constituents have known degrees four and six. Formula (2) follows by partial fractions of (5), or by comparing its seven polynomial coefficients in each residue class modulo 24. These are polynomial identities, not predictions from finite data.

Both displayed generating functions are in lowest terms. The third has nonzero poles of orders three and one at -1 and at primitive third roots. Any quasipolynomial period must therefore be divisible by both two and three. For the fourth, the factors $1+z^4 = \Phi_8(z)$ and $1+z+z^2 = \Phi_3(z)$ do not cancel: the numerator $2z^8 Q_4(z)$ has remainders $2z^3 - 2z$ and -12 modulo these respective factors. Thus any period must be divisible by eight and three. The formulas themselves give periods six and 24, proving minimality. The orders at 1 are five and seven, and yield the stated degrees and leading coefficients.

5 Independent data and scope

The arch transfer computed both diagonals through $n = 10000$ without using either conjectured formula. Exhaustive generation of every noncrossing upper matching gives independent checks for D_3 at $n = 6, \dots, 12$

and for D_4 at $n = 8, \dots, 12$. For each generated matching p , connectivity is checked by the orbit of the permutation $i \mapsto 2n - 1 - p(i)$ in zero-based indices: its orbit from 0 has length n exactly when the two matchings form one loop. The public A400429 b-file agrees throughout its available ranges $n = 6, \dots, 45$ for D_3 and $n = 8, \dots, 45$ for D_4 .

n	$D_3(n)$	$D_4(n)$	n	$D_3(n)$	$D_4(n)$
8	100	42	20	7240	117744
10	340	752	22	10698	215332
12	810	3516	24	15242	371234
14	1610	10948	26	21076	610100
16	2856	27434	28	28420	963652
18	4682	59736	30	37512	1471844

All integers, including omitted indices, are supplied in `diagonal3.csv` and `diagonal4.csv`. The general theorem is an eventual statement; the starting indices 6 and 8 in Theorem 1 follow from the exact generating functions. The general theorem gives a period bound and an effective expression for c_j , rather than a closed formula for the true period or c_j at arbitrary j .

The rainbow representation is classical [1], and the conjecture for D_3 is recorded in A400429 [2]. The D_4 Horner coefficients were supplied by Hunter Hogan in `hogan_diagonal4.py`. The present argument proves these identities by bounding the exposed endpoints when the winding deficit is fixed. The fixed-component expansions in [1, Appendix D] concern a different asymptotic parameter; they are not used to infer either formula here.

References

- [1] P. Di Francesco, O. Golinelli and E. Guitter, *Meanders: A direct enumeration approach*, Nuclear Physics B **482** (1996), 497–535. arXiv:hep-th/9607039.
- [2] H. Hogan, *A400429: semi-meanders by crossings and winding number*, The On-Line Encyclopedia of Integer Sequences, revision 16, 26 September 2026; public text and b-file accessed 6 October 2026. <https://oeis.org/A400429>.